

CENTRAL UNIVERSITY OF KASHMIR

Tulmulla, Ganderbal, Jammu and Kashmir, 191131.



Bachelors in Technology (B Tech)
Computer Science & Engineering (CS&E)

COURSE STRUCTURE

[BoS -14/09/2023]
For 2023 Batch

School of Engineering & Technology
Department of Information Technology

Course Structure for B.Tech. Computer Science & Engineering

Semester I					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BT 101	Mathematics I	3	50	50
2	BT 102	Engineering Physics	3	50	50
3	BT 103	General Chemistry (CHM20-MIN-101)	3	50	50
4	BT 104	Engineering Drawing	2	50	50
5	BT 105	Engineering Workshop	2	50	50
6	BT 107 L	Engineering Physics Lab	1	50	50
7	BT109 L	Chemistry Lab (CHM20-MIN-102)	1	50	50
8	BT 110	Sociology & Elements of Indian History for Engineers	2	50	50
9	BT111	Computer Fundamentals and Applications.	4	50	50
		Total	21		

Semester II					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BT 201	Mathematics II	3	50	50
2	BT 204	Basic Electrical Engineering	3	50	50
3	BT 205	Engineering Mechanics	3	50	50
4	BT 208 L	Basic Electrical Engineering Lab	2	50	50
5	BT 209	Business Communication & Presentation Skills	2	50	50
6	BT 210	Biology	3	50	50
7	BT 213	Computer Programming	4	50	50
8	BT 214 L	Computer Programming Lab	2	50	50
		Total	22		

Course Structure for B.Tech. Computer Science & Engineering

Semester III					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BT 301	Mathematics III (Differential Equation)	3	50	50
2	BTCS 302	Basic Electronics	4	50	50
3	BTCS 303	Data Structures	4	50	50
4	BTCS 308	Database Management Systems	4	50	50
5	BTCS 305 L	Basic Electronics Lab	2	50	50
6	BTCS 306 L	Data Structures Lab	2	50	50
7	BT 309 L	Database Management Systems Lab	2	50	50
8	BT 307	Environmental Studies	2 Audit Non-CGPA	100	
		Total	21		

Semester IV					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BT 401	Mathematics IV (Probability and Statistics)	3	50	50
2	BTCS 408	Design and Analysis of Algorithms	4	50	50
3	BTCS 403	Discrete Structures	4	50	50
4	BTCS 404	Digital Electronics & Logic Design	3	50	50
5	BTCS 406 L	Digital Electronics & Logic Design Lab	2	50	50
6	BT 407	Economics for Engineers	2	50	50
7	BTCS409	Object-oriented Programming	4	50	50
8	BTCS 410	OOP Lab	2	50	50
		Total	24		

Course Structure for B.Tech. Computer Science & Engineering

Semester V					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BTCS 501	Computer Organisation & Architecture	4	50	50
2	BTCS 507	Computer Graphics	3	50	50
3	BTCS 508	Data Communication	4	50	50
4	BTCS 509	Operating Systems	4	50	50
5	BTCS 511 L	Computer Graphics Lab	2	50	50
6	BTCS 512	Web design and development	4	50	50
7	BTCS 513L	Web Design and development lab	2	50	50
		Total	23		

Semester VI					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BTCS 609	Software Analysis & Design	2	50	50
2	BTCS 602	Computer Networks	4	50	50
3	BTCS 604	Theory of Computation	4	50	50
4	BTCS 606 L	Computer Networks Lab	2	50	50
5	BTCS 607	Microprocessor	4	50	50
6	BTCS 608L	Microprocessor Lab	2	50	50
7	BTCS E***	Elective 1	4	50	50
		Total	22		

Course Structure for B.Tech. Computer Science & Engineering

Semester VII					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BTCS 708	Software Engineering	4	50	50
2	BTCS 703	Pre Project/seminar	2	100	
3	BTCS 716	Network Security	4	50	50
4	BTCS 718 L	Network Security Lab	2	50	50
5	BTCS 706	Compiler Design	4	50	50
6	BTCS E***	Elective 2	4	50	50
7	BTCS 707	Industrial Training/Internship	6	100	
		Total	26		

Semester VIII					
S. No.	Course Code	Course Title	Credits	CIA	ESE
1	BTCS 815	Artificial Intelligence	4	50	50
2	BTCS 802	Project	8	50	50
4	BTCS 803	Python Programming	4	50	50
5	BTCS 804	Python Programming Lab	2	50	50
6	BTCS E***	Elective 3	4	50	50
		Total	22		

Total Credits	181		
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Note: Credit requirement for the award of B. Tech Computer Science Engineering =163

Credit Requirement for award of B. Tech Honors Computer Science Engineering =181

Candidate can opt out for electives and 6 credits course/s from VIII semester to be eligible for the award of B. Tech Only

Course Structure for B.Tech. Computer Science & Engineering

List of Electives

S. No.	Course Code	Course Title	Credits
1	BTCS E001	Advanced Algorithms	4
2	BTCS E002	Advanced Computer Architecture	4
3	BTCS E003	Advanced Computer Graphics	4
4	BTCS E004	Advanced Computer Networks	4
5	BTCS E005	Advanced Database Management Systems	4
6	BTCS E006	Advanced Internet Technologies	4
7	BTCS E007	Android Programming	4
8	BTCS E008	Big Data	4
9	BTCS E009	Cloud Computing	4
10	BTCS E010	Computer Vision	4
11	BTCS E011	Cyber Laws and Forensics	4
12	BTCS E012	Digital Signal Processing	4
13	BTCS E013	Distributed and Parallel Computing	4
14	BTCS E014	Embedded Systems	4
15	BTCS E015	Expert Systems	4
16	BTCS E016	Fault Tolerant Computing	4
17	BTCS E017	Graph Theory	4
18	BTCS E018	Green Computing	4
19	BTCS E019	Image Processing	4
20	BTCS E020	Logic Programming	4
21	BTCS E021	Management Information Systems	4
22	BTCS E022	Mobile Computing	4
23	BTCS E023	Neural Networks	4
24	BTCS E024	Next Generation Networks	4
25	BTCS E025	Numerical Methods	4
26	BTCS E026	Pervasive Computing	4
27	BTCS E027	Real Time Operating Systems	4
28	BTCS E028	Visual Programming	4
29	BTCS E029	Simulation and Modelling	4
30	BTCS E030	Software Project Management	4
31	BTCS E031	System Design Using HDL	4
32	BTCS E032	Unix and Shell Programming	4

Course Structure for B.Tech. Computer Science & Engineering

List of Electives			
S. No.	Course Code	Course Title	Credits
33	BTCS E033	Wireless Communication	4
34	BTCS E034	Cryptography & Information Security	4
35	BTCS E035	E-Commerce	4
36	BTCS E036	Enterprise Resource Planning (Erp)	4
37	BTCS E037	Information Retrieval	4
38	BTCS E038	Multimedia	4
39	BTCS E039	Remote Sensing	4
40	BTCS E040	Theory of Languages	4
41	BTCS E041	Data Mining	4
42	BTCS E042	Internet Security and Cyber Law	4
43	BTCS E043	Robotics	4
44	BTCS E044	Introduction to Data Science	4
45	BTCS E045	Computational Data Analytics	4
46		SWAYAM MOOC Course (<i>Department Approved</i>)	4